

ALEXIS PARKER

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SUMMARY OF QUALIFICATIONS

Data scientist and researcher with a Master of Applied Data Science and a BA in Psychology, currently pursuing a PhD in Information Systems and Technology. Experience spans machine learning, AI/NLP systems, behavioral and health research, scientific and survey data, qualitative methods, and interactive applications. Current work includes retrieval-augmented generation testing and source-grounded response evaluation through the AI Squared Student Builder Program, alongside applied research and data products.

EDUCATION

Claremont Graduate University | Claremont, CA Expected May 2030

PhD, Information Systems and Technology | Data Science and Analytics concentration

University of Michigan School of Information | Ann Arbor, MI August 2025

Master of Applied Data Science

California State University, San Bernardino | San Bernardino, CA June 2019

Bachelor of Arts, Psychology

PROFESSIONAL EXPERIENCE

AlphaRose Therapeutics Nov 2025 - Feb 2026

Data Science Intern, AI Interfaces

- Developed tokenization and embedding methods for nucleotide and protein sequences, SMILES strings, and PDB files using RDKit, molecular graphs, and chemoinformatic representations for multimodal drug-design modeling.
- Debugged sequence encoders and intramolecular torsion workflows using RDKit and Snakemake.
- Documented technical methods and workflow decisions for reproducibility and model readiness.

University of Michigan School of Information Jan 2025 - Jan 2026

Graduate Student Researcher

- Conducted literature reviews, qualitative coding, and annotation for health and LLM-focused research.
- Prepared comparative method documentation and communicated methods and findings for technical review.
- Created Python workflows with pandas and scikit-learn and produced KDE, time-series, and regression visualizations for exploratory analysis and research review.
- Contributed to survey research on GenAI adoption among scientists through analysis, methodological review, and communication of development practices.

IHSS Public Authority Apr 2019 - Present

Caregiver

- Coordinate individualized daily support for elderly and disabled clients, including mobility, hygiene, routines, and activities that support independence.

RESEARCH & PROJECT EXPERIENCE

AI Squared Student Builder Program | Project Cerebellum / HISPI 2026 - Present

- Through the Student Builder Program, support Project Cerebellum / HISPI work on unstructured document ingestion, vector stores, embeddings, and retrieval-augmented generation.
- Configure and test parser settings, embeddings, vector-store syncs, and retrieval behavior for a knowledge assistant.
- Review retrieved material and generated responses for relevance, grounding, and accuracy and track issues for continued testing.

[AccessFirst | Los Angeles County Mental Health Resource Navigator](#)

2026

- Co-developed a mental health resource-navigation application using location, insurance category, care type, language, accessibility needs, and telehealth preference to narrow available resources.
- Designed a structured conversational workflow that supports follow-up refinement while keeping resource retrieval tied to the user's stated needs and available source data.
- Defined privacy, crisis-routing, and scope guidance that keeps the application focused on navigation rather than diagnosis, treatment recommendations, appointment booking, or insurance verification.
- Team received Second Prize in the 2026 China-U.S. Young Maker Competition.

[Depression Risk Modeling | NHANES](#)

2025

Graduate Project

- Outer-merged seven NHANES modules and applied adult, interview, examination, and PHQ validity filters to create a 467-record depression-severity modeling table.
- Compared classification approaches and evaluated performance using confusion matrices, ROC-AUC, hyperparameter tuning, and SHAP.

[Depression Index Study | CDC BRFSS](#)

2025

Graduate Project

- Analyzed 2022 BRFSS data with prepared stages above 400,000 records and constructed a seven-component exploratory index using mental-health, physical-health, social, stress, life-satisfaction, and depression-diagnosis measures.
- Examined state and demographic variation and assessed how missing-data and index-construction choices affect interpretation.

California State University, San Bernardino

Apr 2019 - Jun 2019

Behavioral Neuroscience Research Assistant

- Conducted a controlled behavioral study examining substance-related conditioned preference in adolescent rats using structured research protocols.
- Recorded laboratory observations and analyzed study results using three-way ANOVA.

PRESENTATIONS | CONFERENCES

[North American Cartographic Information Society \(NACIS\) Annual Meeting](#)

October 2026

Scheduled Co-Presenter

Communicating Healthcare Affordability Through Maps: A Visual Analytics Framework Using NHIS and MEPS Data

Sandipta Khare, Alexis Parker, and Hengwei Zhang

- Co-authored research using 2023 NHIS and MEPS public-use data to examine cost-related barriers to mental health counseling and therapy and annual healthcare expenditure burden.
- Used regional mapping, visual analytics, and uncertainty-aware cartographic design to communicate affordability patterns while acknowledging geographic limitations.

PUBLICATIONS

O'Brien, G., Parker, A., Eisty, N., & Carver, J. (2025, revised 2026). [A survey of generative AI adoption and perceived productivity among scientists who program](#). arXiv:2512.19644v3.

- Co-authored a survey of 868 scientists who program examining GenAI adoption, tool preferences, development practices, and perceived productivity.

TECHNICAL SKILLS

Programming, AI & Tools: Python, SQL, R, pandas, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow, NLP/LLMs, RAG, embeddings, SHAP, Git/GitHub, Jupyter, Google Cloud Platform, Snakemake, RDKit, Streamlit

Research & Evaluation: Qualitative coding, annotation, retrieval testing, experimental design, survey research, model evaluation, classification, regression, clustering